Combinations

1 Evaluate the following expressions:

a $5!$

b $4!6!$

c $\frac{11!}{5!6!}$

d $\frac{19!}{15!4!}$

e ${}_7C_2$

f $\binom{7}{3}$

g $\binom{10}{9}$

h $\frac{{}_{10}C_5}{{}_{10}C_6}$

2 How many different unordered selections can be made from a set of 7 elements?

3 How many different unordered selections can be made from the set of letters of the Greek alphabet, which contains 24 letters?

4 Determine the number of ways 3 cards can be chosen from a standard deck of 52 cards if:

a Exactly one of the cards is to be a Queen.

b None of the cards is to be a Queen.

c At most one of the cards is to be a Queen.

5 Amelia is playing a game of cards. She will randomly draw 5 cards from a standard deck of 52 cards, and wants only 3 of the cards to have clubs on them. How many different selections would contain the cards she wants?

6 Determine the number of ways 6 people can be chosen from a group of 11 people if:

a The eldest person is included.

b The eldest person is not included.

c The eldest and youngest people are both to be included.

Probability with combinations

7 There are 5 red, 5 white and 3 blue marbles in a bag. 5 marbles are chosen at random from the bag, without replacement. Find the probability that:

a The selection consists of all red marbles.

b The selection consists of 4 red and 1 white marble.

c There is at least 1 marble of each color.

d There are no red marbles.

8 5 people are to be selected from a larger group of 10 candidates. If Amelia is among the candidates, find the probability that she will be among those selected.

9 5 cards are chosen at random without replacement from a standard deck of 52 cards. Find the probability that they are all red.

10 3 letters of the word ENVIOUS are chosen at random. Find the probability that the selection includes just 1 vowel.

11 6 cards are chosen from a standard deck of 52 cards, without replacement.

- **a** Find the probability that they are all of the same suit.
- **b** Find the probability that there will be at least one of each suit.

12 A box contains 6 pens of different colors: red, green, blue, yellow, black and white. Two pens are drawn at random without replacement.

- **a** How many possible selections are there?
- **b** Find the probability of drawing the green and black pens.

13 A container contains twelve cards. The cards are marked with the numbers 1 to 12. From the container, five cards are drawn at random without replacement.

- **a** How many possible selections are there?
- **b** Find the probability of selecting 2 as the first number.
- **c** Find the probability of selecting numbers 2, 1 and 4 (in any order) as the first three of the 5 cards.

Applications

14 From 8 teachers and 7 students, 4 people are randomly selected.

- **a** Find the probability that a particular student is chosen.
- **b** Find the probability that there are at least three teachers in the selection.

15 A drawer contains 18 batteries, of which 4 h[redacted]w charge and the rest are fully charged. Vanessa needs to choose three of the batteries to operate a remote control. Find the probability that:

a All of the batteries she chooses are fully charged.

b At least one of the batteries she chooses has low charge.

16 An airline wants to allocate 5 flight attendants to a particular flight route. They randomly select them from 8 staff who speak a European language only and 6 staff who speak an Asian language only.

a Find the probability that all the flight attendants speak a European language only.

b Find the probability that there are more Asian than European language speaking attendants.

17 In court, a jury panel of 8 members is to be made up from a group of 20 candidates. In how many different ways can a panel be formed?

18 At a film festival, 6 films are to be chosen to go in the running for grand prize of best film. There are 8 foreign films and 7 local films to choose from. If the selection is made randomly, find the probability that 4 foreign films and 2 local films are chosen.

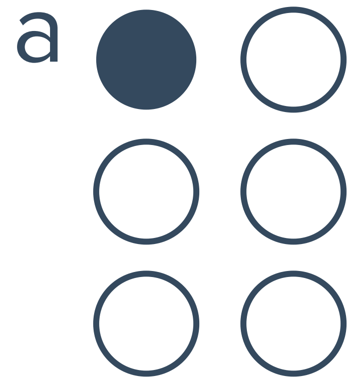
19 In a bicycle race, a quinella is a bet on the first 2 bicycles that finish the race, but the order in which these 2 bicycles place does not matter. How many different quinella bets are possible for a bicycle race where 14 bicycles are competing?

20 In a game of football, each team must have 1 goalkeeper and 10 other players on the field. A football squad consists of 17 players, 2 of which are goalkeepers. In how many ways can players be chosen to start the game?

21 A tea shop sells 10 different types of tea and 7 different sets of teacups, and has room to display 6 types of tea and 3 sets of teacups. The owner changes the display each month. How many different display combinations of tea and tea cups are possible?

22 A teacher wants to divide her 12 students into a group of 4, then a group of 5 and finally a group of 3. In how many ways can she do this?

23 In the Braille system of writing, six dots are arranged, and a particular combination of raised dots represents a particular letter. The letter 'a' is represented by one dot being raised, as shown:



a In how many ways can two dots be raised?

b In how many ways can 1 to 6 dots be raised?

24 In a Lotto draw, 5 different numbers are chosen from the numbers 1 to 35. How many possible selections are there? Assume that order does not matter.

25 Frank's organization wants to form a partnership with a charity. After a day of speaking with representatives from different charities, he looks in his wallet and sees that he has 3 identical business cards from one charity and one business card from each of the other 9 charities he spoke with. If he wants to select 3 cards to keep, no two from the same charity, how many different selections can he make?

26 Four cards of different suits (diamonds, hearts, spades and clubs) are placed in a pile. A card is selected from the pile. Its suit is recorded and the card is returned to the pile. A second card is then chosen and its suit is also recorded.

a How many possible outcomes are there?

b Find the probability of drawing two diamonds.

c Find the probability that the first card is a heart and the second card is a club.

d Find the probability of both cards being black.

27 A newspaper editor is deciding which of 5 articles to print on the front page.

a If she can only choose 2 of them for the front page, how many different selections are possible?

b If Jack wrote one of the 5 articles, find the probability that his article is chosen for the front page.

28 A selection of 4 people are to be chosen from a group of 7 people. How many selections are possible if the youngest or oldest is included but not both?

29 A team of 3 is to be chosen at random from a group of 5 girls and 6 boys. Determine the number of ways the team can be chosen if:

a There are no restrictions.

b There must be more boys than girls.

30 The state government needs to determine how school funding is allocated for the following year. Of the 8 public schools that will be affected, 5 of them are asked to attend a meeting. Determine the number of ways the 5 representative schools can be chosen if:

a The two least funded schools must attend.

b Neither of the two least funded schools can attend.

c Saint Hibiscus Girls' and Saint Hibiscus Boys' are two of the public schools affected, and exactly one of them must be chosen.

31 In a football squad, there are 9 midfielders. 5 midfielders are selected to play in a game.

a How many selections of 5 midfielders are possible?

b In how many of these selections are Frank and Glen included?

c Find the probability that Frank and Glen will be included in the selection.

32 Companies prefer their product to be placed on the middle shelf in supermarkets. There are 6 brands, but only space for 3 of them on the middle shelf.

a In how many ways can the products for the middle shelf be selected?

b Find the probability that a particular brand is placed on the middle shelf.

33 5 boys and 6 girls are part of the debate team. 4 of them must represent the school at the upcoming debate tournament. Determine the number of ways the team of 4 can be formed if:

a It must contain 2 boys and 2 girls.

b It must contain at least 1 boy and 1 girl.

34 A company has recently expanded overseas and is looking to recruit 5 new Resource Analysts. They have narrowed it down to 8 equally qualified applicants, of which 2 of them can speak a second language. In how many ways can they fill the positions if exactly one of the recruits must speak a second language?

35 In a court case, the defendant can choose a jury of 9 members from 15 candidates consisting of 9 women and 6 men. They notice that one particular lady looks ill and wish to avoid choosing her. How many different panels of jury members can they create if they want at least 7 women to make up the jury?

36 In a radio discussion about an upcoming election, 7 Democrat voters, 5 Republican voters and 5 Independent voters called up to have a say. The radio announcers want to choose 1 Democrat voter, at least 4 Republican voters and at least 3 Independent voters to talk to on air. In how many ways can they select the callers?

37 A senior science teacher wants to receive feedback from a selection of students on whether they thought the exam was fair. The table shows the range of scores and the number of students who fell within each range:

Score range	$0-40$	$41-50$	$51-60$	$61-70$	$71-80$	$81-100$
Number of students	2	4	5	7	10	5

She selects 2 students to provide feedback, but does not want more than 1 student from any score range. In how many ways can she select the 2 students?

38 Quiana is ordering a salad at a restaurant. She can order a salad consisting of just lettuce, or she can also add any of the following ingredients:

$$\text{Onions, Cucumbers, Tomatos, Cheese, Olives, Carrots, Mushrooms}$$

How many different variations of a salad can she possibly choose?

39 A local grocery store offers 26 different flavors of ice-cream. A cone can hold 2 flavors.

a How many selections of two flavors are possible?

b If two of the available flavors are chocolate and vanilla, find the probability of selecting both chocolate and vanilla.

40 You order 12 sandwiches for your office lunch. 5 are roast beef, 5 are ham and 2 are turkey. The sandwich shop wrapped them but forgot to label which meat was in each sandwich. If you pick 4 sandwiches at random, find the probability of the following as a percentage to two decimal places.

a All 4 sandwiches are roast beef.

b Exactly one sandwich is ham.

c 2 are ham and 2 are turkey.

41 4 cards are chosen at random without replacement from a standard deck of 52 cards. Find the following probabilities, as percentages to two decimal places:

a That they are all black.

b That you get two kings and two queens.

c That you get three kings and one random non-king card.

42 Bottles of water are capped by machinery on a production line. After they have been capped, bottles are randomly selected to test that they are correctly sealed. Of the first 20 bottles to come off the production line, 7 of them are not correctly sealed. Georgia randomly selects 5 of the first 20 bottles.

a Find the probability that all of the ones she selected are properly sealed.

b Find the probability that at least one of the bottles she selected is not properly sealed.